Can LLM Judgment Capture Human Preference?

An Alignment Study Between Automated and Human Evaluation of LLM Task Performance

Yifan Liu
Student #1007341681

Department of Mechanical and Industrial Engineering
University of Toronto

March 2026

Outline

1. Research Problem
2. Method: LLM-as-Judge
3. Dataset: Chatbot Arena
4. Results
5. Summary & Ongoing Work

The Research Problem

Evaluating LLMs is expensive. **Human evaluation** (crowdsourced voting) is the gold standard — but slow and costly to scale.

A popular alternative: use a **strong LLM as a judge** to evaluate other LLMs automatically.

Research Question

Can an LLM judge reliably capture human preferences when evaluating LLM task performance?

Existing work reports high overall agreement ($r \geq 0.94$) between automated LLM judges and human voters.

But is this agreement uniform? Or does the LLM judge agree with humans on some tasks and fail on others?

Why This Matters

1. **Replacing human evaluation**
   If LLM judges fail on certain task types, using them as a blanket replacement for human evaluation is unreliable.

2. **Model development**
   RLHF, DPO, and other alignment methods use automated scores as training signals. If the automated judge disagrees with humans on math or translation, models trained on those signals may be optimized for the wrong objective.

3. **Model selection**
   Practitioners choose LLMs based on benchmark leaderboards. If the leaderboard is driven by an LLM judge that doesn't match human preferences for their use case, they may pick the wrong model.

LLM-as-Judge: How It Works

The **LLM-as-judge** paradigm uses a strong LLM (GPT-4-turbo) to automatically evaluate other models' outputs. **WildBench** (Lin et al., ICLR 2025 Spotlight) is a leading implementation:

1. **Input**: A real-user prompt (from WildChat) + the target model's response
2. **Checklist generation**: GPT-4-Turbo and Claude-3-Opus jointly generate a **task-specific checklist** of 5–10 verifiable questions tailored to the prompt (e.g., "Does the code handle edge cases?" for a coding task)
3. **Evaluation**: The LLM judge evaluates the response against the checklist using Chain-of-Thought reasoning, producing a structured analysis of strengths and weaknesses before arriving at a final judgment

Why Checklists?

Without checklists, WB-Score achieves $r = 0.905$ with human Elo. **With checklists, it rises to** $r = 0.925$ (ablation study, Section 4.3). Checklists reduce subjectivity by grounding evaluation in concrete, task-specific criteria.

1,024 tasks sourced from WildChat · 12 task categories · 60+ models evaluated

Two Metrics: WB-Score and WB-Reward

WB-Score (Pointwise)

- Judge rates **one response** on a 1–10 scale against the checklist
- Score rubric:
  - 1–2: Makes no sense
  - 3–4: Poor, doesn't help user
  - 5–6: Fair but has issues (errors, hallucinations)
  - 7–8: Good but could improve
  - 9–10: Perfect, fully solves the problem
- Rescaled: $S' = (S - 5) \times 2$
- Cheaper ($5, 10 min per model)
- Achieves $r = 0.955$ with Arena Elo (top models)

WB-Reward (Pairwise)

- Judge **compares** test model vs. a baseline model using the checklist
- 5 outcome levels, scored as:
  - Much better: $+1$
  - Slightly better: $+0.5$
  - Tie: 0
  - Slightly worse: -0.5
  - Much worse: -1
- **3 baselines**: GPT-4-Turbo, Claude-3-Haiku, Llama-2-70B-chat (averaged)
- **Length penalty**: if winner is $> K$ chars longer, slight win $\rightarrow$ tie ($K = 500$)
- Achieves $r = 0.984$ with Arena Elo (top models)

Task-Specific Checklists: Example

Example prompt (G20 essay task from WildBench)

"Write an essay on the impact of the G20 summit on the global economy... it has to have more than 1200 words. Write it beautiful and poetic. Use extensive vocabulary... Use some ancient Indian historical references."

Generated checklist (5–10 questions per task):

1. Does the essay contain more than 1200 words as requested?
2. Is the language beautiful and poetic, with extensive vocabulary?
3. Does it include factual data on the G20's economic impact?
4. Are there references to young people shaping the world?
5. Does it include ancient Indian historical references?
6. Is the essay structured in a clear and logical manner?

The judge then evaluates each model's response against **these same questions**, ensuring consistent, interpretable, and comparable scoring across all models. Checklists are generated once per task and reused for every model.

Chatbot Arena: Human Preference at Scale

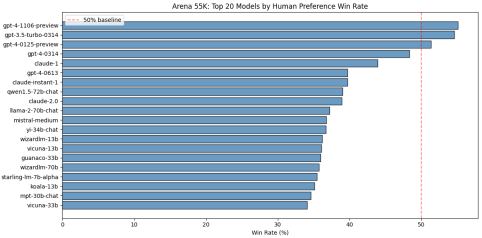
Chiang et al., ICML 2024

The largest open human preference dataset for LLM evaluation:

- A user submits a prompt and sees responses from **two anonymous LLMs** side by side
- The user votes for the response they prefer
- **57,477 pairwise battles**, 64 models
- Full text of prompts and responses available
- Supplemented by **1.7M voting records**

Rankings computed via **Bradley-Terry model** → Elo ratings (same concept as chess). Widely considered the **gold standard** for alignment evaluation.

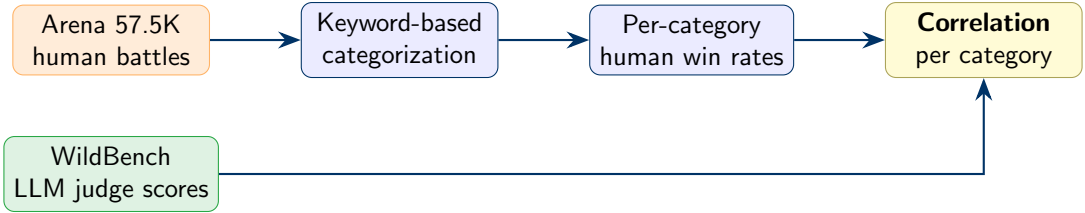
This is our ground truth: what do humans actually prefer?



Top 20 models by Elo (Arena 55K, Bradley-Terry with bootstrap).

Applying Task Categorization to Arena Data

Arena provides aggregate Elo rankings but **no task-category breakdown**. We add one:



- Classify each Arena prompt into WildBench's 12 task types via keyword heuristics (e.g., "code", "debug", "Python" → Coding & Debugging)
- Compute per-model human win rates within each category
- Compare against WildBench's LLM judge scores for **12 overlapping models**
- All processed with PySpark (∼1.8M records). **No LLM inference** — all judgments precomputed.

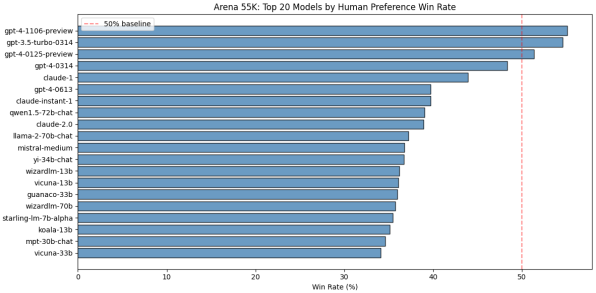
Result 1: Overall — the LLM Judge Agrees with Humans

Cross-benchmark correlation (12 overlapping models):

	Pearson	Spearman
Elo vs WB-Score	**0.954**	0.923
Elo vs WB-Reward	**0.943**	0.895

High overall agreement ($p < 10^{-5}$), partly driven by large inter-model gaps — every benchmark can tell GPT-4 from Llama-2.

This confirms previous findings. But the story changes when we look per category...



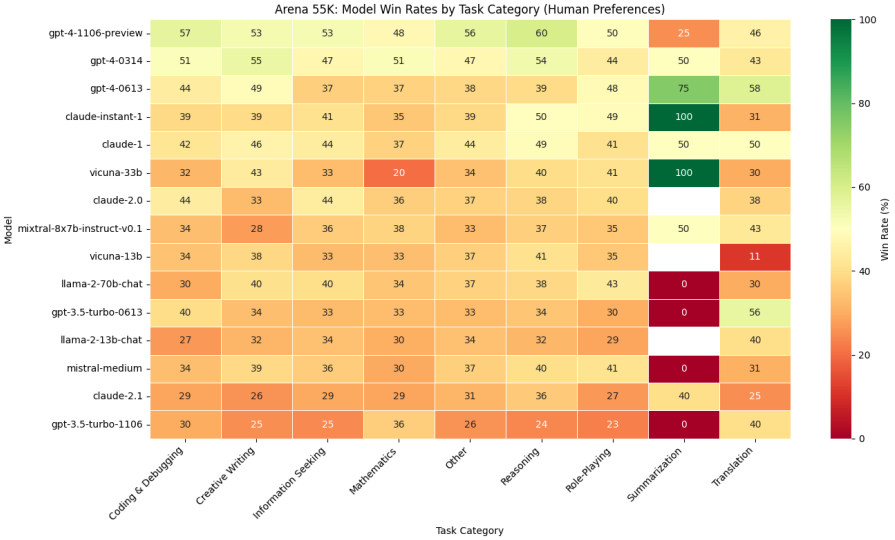
Top 20 models by human preference (Arena 55K, Bradley-Terry Elo)

Result 2: Per Category — Agreement Varies Dramatically

Task Category	N	Pearson (WB-Score)	Pearson (WB-Reward)
Reasoning	11	**0.749**	0.693
Creative Writing	11	0.713	0.649
Coding & Debugging	12	0.689	0.661
Role-Playing	11	0.689	0.642
Information Seeking	12	0.595	0.554
Mathematics	11	0.558	0.581
Translation	5	**0.146**	0.144
Overall	12	*0.954*	*0.943*

The 0.95 overall correlation **masks a range from 0.75 to 0.15**. The LLM judge captures human preferences well for reasoning, but **poorly for translation and mathematics**.

Result 3: Human Rankings Shift Across Categories



Result 3 (cont'd): Category-Specific Leaders

Task Category	Top Model (human votes)	Win Rate
Coding & Debugging	gpt-3.5-turbo-0314	61.2%
Creative Writing	gpt-3.5-turbo-0314	61.7%
Information Seeking	gpt-3.5-turbo-0314	60.0%
Reasoning	**gpt-4-1106-preview**	60.2%
Mathematics	gpt-3.5-turbo-0314	75.0%
Role-Playing	mpt-30b-chat	70.0%
Translation	gpt-4-0613	58.3%

Implication

A single leaderboard ranking — whether from an LLM judge or human Elo — does not tell the full story. The "best" model depends on the task. An LLM judge that agrees with humans overall may give **misleading guidance** for specific use cases.

Summary

Research question: Can an LLM judge reliably capture human preferences when evaluating LLM task performance?

Answer: *It depends on the task.*

1. **Overall**: Yes — LLM judge scores correlate strongly with human Elo ($r = 0.95$).
2. **Per category**: Agreement ranges from $r = 0.75$ (reasoning) to $r = 0.15$ (translation). The LLM judge is a reliable proxy for some tasks but not others.
3. **Rankings shift**: The best model overall is not the best in most categories. Aggregate scores — from either LLM judges or human Elo — hide task-specific strengths and weaknesses.

Takeaway: High overall correlation between LLM judges and humans should not be taken as blanket validation. Per-category analysis is necessary.

Ongoing Work

Completed:

- ✓ Arena 57.5K: task categorization, per-category human win rates
- ✓ Elo ratings via Bradley-Terry model
- ✓ Cross-benchmark correlation — overall and per category (7 categories)
- ✓ 7 visualizations: heatmaps, scatter plots, bar charts

In progress:

1. Per-category Elo (fairer than win rate for uneven matchup pools)
2. 1.7M Arena voting records for more robust Elo estimates
3. AlpacaEval 2 integration for three-way comparison
4. Bootstrap confidence intervals for correlation estimates
5. Improved categorization coverage (currently 40% of prompts classified)
6. Full PySpark pipeline at scale

Thank You

Questions?

References:

Lin et al., "WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild," ICLR 2025

Chiang et al., "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference," ICML 2024

Dubois et al., "Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators," COLM 2024